QUANTUM SECURITY INTELLIGENCE

Security & Post-Quantum Readiness Audit Report

fabric-lib-go @ f8b4f4a833e4 · Hyperledger Fabric (LF Decentralized Trust) — public repository (passive review)

Organization	QSI
Report date	2026-11-06 10:15 UTC
Report version	1
Audit run	9f3b8d12-9473-49de-9ebb-9f2f640f8653 · Document audit · 2026-11-06 10:15 UTC
Analyzer engine	qsi-engine 1.3.0
Risk profile	Crypto exchange, custodian or wallet
Content SHA-256	`967ff3502b91c3f489ae4dc96bfd80d7f227c73a45038ab60ffb4c654876bd06`
Signature key (Ed25519, SHA-256 fingerprint; the signature is the separate file report.sig.json)	`5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`

Contents

1. Executive summary
2. Scope and boundaries
3. Methodology
4. Assets examined and not examined
5. Architecture overview
6. Overall security posture
7. Findings by severity
8. Detailed findings
9. Evidence
10. Post-quantum assessment
11. Limitations and assumptions
12. Remediation plan
13. Priorities for 30, 60 and 90 days
14. Verification after remediation
15. Tools used
16. Sources and normative documents
17. Appendices

This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.

Executive summary

11	0	1	3	0
Findings in total	Critical	High	Medium	Low

Classification	Count
Confirmed technical vulnerability	0
Known vulnerable component	8
Confirmed configuration error	0
Potential cryptographic risk	0
Long-term post-quantum risk	3
Improvement recommendation	0
Insufficient data	0

Top 5

ID	Severity	Detailed findings	Priority score
QSI-2026-17514	High	google.golang.org/grpc v1.83.1 is affected by GHSA-2v4p-qf9q-27wj (CVE-2026-84445, GO-2026-6443)	50.2
QSI-2026-17511	Medium	Quantum-vulnerable ECDSA: role not determined (10 file(s)) preliminary	34.2
QSI-2026-17512	Medium	Quantum-vulnerable Ed25519: role not determined (6 file(s)) preliminary	34.2
QSI-2026-17513	Medium	Quantum-vulnerable RSA: role not determined (6 file(s)) preliminary	34.2
QSI-2026-17515	Info	golang.org/x/net v0.58.0 is affected by GO-2026-6603 (CVE-2026-78659) (no severity rating) preliminary	17.8

Mosca's inequality (x + y > z): x = 10, y = 5, z = 8 → migration would finish too late under this scenario. z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography. Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Scope and boundaries

Mode: Document audit

Repository: https://github.com/hyperledger/fabric-lib-go @ f8b4f4a833e4140a4d006fc2e0a46823952b5f1c — evidence links open the lines at this commit.

Supplied inputs

67 files, 263295 bytes in total, 0 redaction(s); the per-file SHA-256 list is in appendix C.

Methodology

The audit was performed by the QSI engine 1.3.0. Each analyzer emits facts (rule, parameters, evidence); finding texts are generated from the rule catalog. Methodology sources are listed under "Sources and normative documents".

Prioritization uses an explainable weighted model (severity, exploitability, exposure, confidentiality/integrity/availability impact weighted by the "Crypto exchange, custodian or wallet" profile, asset criticality, urgency, ease of remediation) with a confidence multiplier. CVSS v3.1 is applied only to technical vulnerabilities with a defined vector. Long-term quantum risks use Mosca's inequality; their severity is capped at Medium and they are never presented as exploitable vulnerabilities.

Finding classes: confirmed technical vulnerability; known vulnerable component (a dependency version named in an advisory; whether the affected code is reachable is assessed separately, and a component confirmed by review becomes a vulnerability); confirmed configuration error; potential cryptographic risk; long-term post-quantum risk; improvement recommendation; insufficient data. Heuristic findings are marked preliminary; analyst decisions are shown next to the findings.

Coverage of checks

Area	Status	Count	Examples
source_code	checked	65	bccsp/aesopts.go: 82 lines scanned with 16 rule patterns and 266 registry patterns; bccsp/bccsp.go: 134 lines scanned with 16 rule patterns and 266 registry patterns; bccsp/ecdsaopts.go: 49 lines scanned with 16 rule patterns and 266 registry patterns
dependencies	checked	2	go.mod: 41 dependencies parsed (41 pinned); tools/go.mod: 5 dependencies parsed (5 pinned)
entropy	checked	1	random-number generation: 65 code files read with 0 macro definitions from the tree; 0 ineffective guard(s), 0 weak random-bytes provider(s), 0 weak key-generation site(s)
osv_advisories	checked	1	42 pinned packages queried at OSV.dev; 2 affected

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

Assets examined and not examined

Supplied inputs: 69 (input files: C)

Architecture overview

Static review of the source tree fabric-lib-go at f8b4f4a833e4140a4d006fc2e0a46823952b5f1c. 67 files analysed; of 150 files, 63 are test, fuzzing, benchmark or example code and were left out, 20 are not read by any analyzer, 0 exceed the per-file limit and 0 did not fit the total limit. Files no analyzer reads include: .github/settings.yml, .github/workflows/build.yml, .github/workflows/schedule.yml, .gitignore, CODEOWNERS. 1581 files in build-output or dependency-cache directories (.git, .hg, .venv, __pycache__, build, dist, node_modules, target, vendor, venv) were not read. No requests were sent to the owner's systems.

Overall security posture

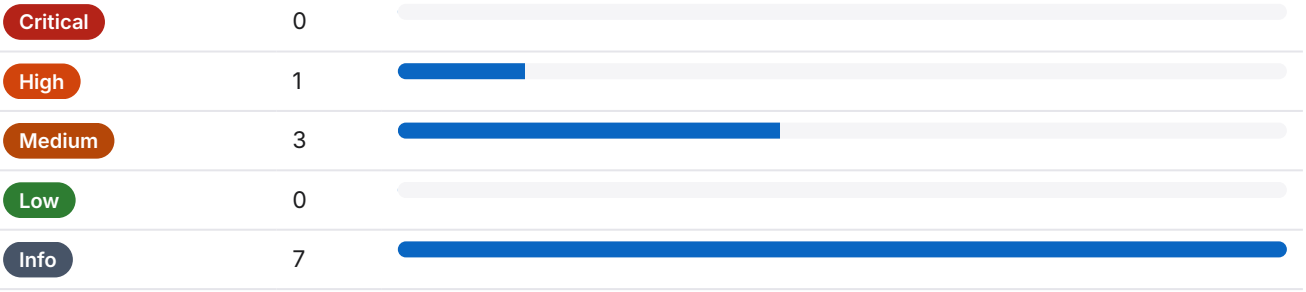
88
Indicative posture index (0–100)

64.9%
Quantum-vulnerable share of inventory

37
Cryptographic inventory

Derived from findings in scope (confirmed and potential issues weighted by severity). Indicative only; not comparable across differently scoped audits.

Findings by severity



ID	Detailed findings	Classification	Severity	Confidence
QSI-2026-17514	google.golang.org/grpc v1.83.1 is affected by GHSA-2v4p-qf9q-27wj (CVE-2026-84445, GO-2026-6443)	Known vulnerable component	High	Medium
QSI-2026-17511	Quantum-vulnerable ECDSA: role not determined (10 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-17512	Quantum-vulnerable Ed25519: role not determined (6 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-17513	Quantum-vulnerable RSA: role not determined (6 file(s))	Long-term post-quantum risk	Medium	Medium, preliminary
QSI-2026-17515	golang.org/x/net v0.58.0 is affected by GO-2026-6603 (CVE-2026-78659) (no severity rating)	Known vulnerable component	Info	Low, preliminary
QSI-2026-17516	golang.org/x/net v0.58.0 is affected by GO-2026-6610 (CVE-2026-78660) (no severity rating)	Known vulnerable component	Info	Low, preliminary
QSI-2026-17517	golang.org/x/net v0.58.0 is affected by GO-2026-6611 (CVE-2026-78669) (no severity rating)	Known vulnerable component	Info	Low, preliminary
QSI-2026-17518	golang.org/x/net v0.58.0 is affected by GO-2026-6612 (CVE-2026-78663) (no severity rating)	Known vulnerable component	Info	Low, preliminary
QSI-2026-17519	golang.org/x/net v0.58.0 is affected by GO-2026-6617 (CVE-2026-97032) (no severity rating)	Known vulnerable component	Info	Low, preliminary
QSI-2026-17520	golang.org/x/net v0.58.0 is affected by GO-2026-6860 (CVE-2026-78661) (no severity rating)	Known vulnerable component	Info	Low, preliminary
QSI-2026-17521	golang.org/x/net v0.58.0 is affected by GO-2026-6866 (CVE-2026-97045) (no severity rating)	Known vulnerable component	Info	Low, preliminary

Detailed findings

QSI-2026-17514 · DEP-KNOWN-ADVISORY

High

google.golang.org/grpc v1.83.1 is affected by GHSA-2v4p-qf9q-27wj (CVE-2026-84445, GO-2026-6443)

Known vulnerable component | Confidence: Medium

OSV.dev reports GHSA-2v4p-qf9q-27wj (CVE-2026-84445, GO-2026-6443) for Go package google.golang.org/grpc at version v1.83.1: gRPC-Go xDS servers: Denial of Service (DoS) via crash due to missing `:authority` and `Host` headers. The advisory lists only unexported functions (google.golang.org/grpc/internal/transport.http2Server.HandleStreams, google.golang.org/grpc/internal/transport.http2Server.operateHeaders, google.golang.org/grpc/internal/xds/server.RouteAndProcess), which code outside the package cannot call by name, so reachability was not determined.

Affected asset	dependency:Go/google.golang.org/grpc@v1.83.1
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 10:15 UTC
Priority score	50.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Depends on whether the vulnerable code path is reachable in your application.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GHSA-2v4p-qf9q-27wj
```

EVIDENCE

tool_output · OSV.dev GHSA-2v4p-qf9q-27wj · line 19 · SHA-256 3566bf539cd1bed7...


```
{"id": "GHSA-2v4p-qf9q-27wj", "aliases": ["CVE-2026-84445", "GO-2026-6443"], "modified":
"2026-09-25T18:00:05.270054427Z", "summary": "gRPC-Go xDS servers: Denial of Service (DoS) via crash
due to missing `:authority` and `Host` headers", "severity": [{"type": "CVSS_V4", "score":
"CVSS:4.0/AV:N/AC:L/AT:N/PR:N/UI:N/VC:N/VI:N/VA:H/SC:N/SI:N/SA:N/E:X/CR:X/IR:X/AR:X/MAV:X/MAC:X/
MAT:X/MPR:X/MUI:X/MVC:X/MVI:X/MVA:X/MSC:X/MSI:X/MSA:X/S:X/AU:X/R:X/V:X/RE:X/U:X"}],
"declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed. A Go function counts only in a file that also names its package (a method: its type); unexported functions only through the exported ones.

RECOMMENDATION

Upgrade to a fixed version (1.82.2, 1.83.2, 1.84.0-dev.0.20260825144003-d5a41119e0e3, 1.85.0-dev.0.20260825072537-93e31b48545e).

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GHSA-2v4p-qf9q-27wj

QSI-2026-17511 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable ECDSA: role not determined (10 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

14 reference(s) to ECDSA across 10 file(s), e.g. bccsp/pkcs11/ecdsakey.go:19. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:ECDSA:general
Data source	bccsp/pkcs11/ecdsakey.go
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 10:15 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open bccsp/pkcs11/ecdsakey.go:19 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · bccsp/pkcs11/ecdsakey.go · line 19 · SHA-256 87680e3cf321a1d9… · File 41b35f8b4e01edc5…

```
18 | import (
19 | "crypto/ecdsa"
20 | "crypto/x509"
```

file_excerpt · bccsp/pkcs11/pkcs11.go · line 11 · SHA-256 8da8b0c88ebe0194… · File 98a15788cd8cdca1…

```
10 | "context"
11 | "crypto/ecdsa"
12 | "crypto/elliptic"
```

file_excerpt · bccsp/pkcs11/pkcs11.go · line 330 · SHA-256 c46c43f1ab599be0… · File 98a15788cd8cdca1…

```
329 | if csp.softVerify {
330 |         return ecdsa.Verify(k.pub, digest, r, s), nil
331 | }
```

file_excerpt · bccsp/sw/ecdsa.go · line 19 · SHA-256 1515eca1cc78dc2b… · File f0d2594a515e856b…

```
18 | import (
19 | "crypto/ecdsa"
20 | "crypto/rand"
```

file_excerpt · bccsp/sw/ecdsa.go · line 28 · SHA-256 5e03fc2ced291506… · File f0d2594a515e856b…

```
27 | func signECDSA(k *ecdsa.PrivateKey, digest []byte, opts bccsp.SignerOpts) ([]byte, error) {
28 | r, s, err := ecdsa.Sign(rand.Reader, k, digest)
29 | if err != nil {
```

file_excerpt · bccsp/sw/ecdsa.go · line 56 · SHA-256 cc80b9b87564d0ae… · File f0d2594a515e856b…

```
55 |
56 | return ecdsa.Verify(k, digest, r, s), nil
57 | }
```

file_excerpt · bccsp/sw/ecdsakey.go · line 19 · SHA-256 da30811132376f30… · File 194f71f7e0d952ec…

```
18 | import (
19 | "crypto/ecdsa"
20 | "crypto/elliptic"
```

file_excerpt · bccsp/sw/fileks.go · line 11 · SHA-256 e5b44015f00ef128… · File a0c92a6cfdcba39b…

```
10 | "bytes"
11 | "crypto/ecdsa"
12 | "crypto/ed25519"
```

file_excerpt · bccsp/sw/keyderiv.go · line 20 · SHA-256 9f85b73ac6ec1422… · File fb6673a63d52fa58…

```
19 | import (
20 | "crypto/ecdsa"
21 | "crypto/hmac"
```

file_excerpt · bccsp/sw/keygen.go · line 10 · SHA-256 7983932d009be78f… · File 51177e8239b008e1…

```
 9 | import (
10 | "crypto/ecdsa"
11 | "crypto/ed25519"
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17512 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable Ed25519: role not determined (6 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

39 reference(s) to Ed25519 across 6 file(s), e.g. bccsp/sw/ed25519.go:15. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:Ed25519:general
Data source	bccsp/sw/ed25519.go
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 10:15 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open bccsp/sw/ed25519.go:15 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · bccsp/sw/ed25519.go · line 15 · SHA-256 be962610d24d0e25… · File 071a175c9ce2398e…

```
   14 | func signED25519(k *ed25519.PrivateKey, msg []byte, opts bccsp.SignerOpts) ([]byte, error) {
   15 | signature := ed25519.Sign(*k, msg)
   16 | return signature, nil
```

file_excerpt · bccsp/sw/ed25519.go · line 20 · SHA-256 aa94dd71ac249c87… · File 071a175c9ce2398e…

```
   19 | func verifyED25519(k *ed25519.PublicKey, signature, msg []byte, opts bccsp.SignerOpts)
(bool, error) {
   20 | return ed25519.Verify(*k, msg, signature), nil
   21 | }
```

file_excerpt · bccsp/sw/ed25519.go · line 23 · SHA-256 6f33081caebdef55… · File 071a175c9ce2398e…

```
   22 |
   23 | type ed25519Signer struct{}
   24 |
```

file_excerpt · bccsp/sw/ed25519key.go · line 18 · SHA-256 c161f2e6d9d89ebe… · File 5fd0b8364abab17d…

```
   17 |
   18 | type ed25519PrivateKey struct {
   19 | privKey *ed25519.PrivateKey
```

file_excerpt · bccsp/sw/ed25519key.go · line 24 · SHA-256 ee921a0dccc1f590… · File 5fd0b8364abab17d…

```
23 | // if this operation is allowed.
24 | func (k *ed25519PrivateKey) Bytes() ([]byte, error) {
25 | return nil, errors.New("Not supported.")
```

file_excerpt · bccsp/sw/ed25519key.go · line 29 · SHA-256 488f2f75c771d7f1... · File 5fd0b8364abab17d...

```
28 | // SKI returns the subject key identifier of this key.
29 | func (k *ed25519PrivateKey) SKI() []byte {
30 | if k.privKey == nil {
```

file_excerpt · bccsp/sw/fileks.go · line 147 · SHA-256 1cfe306c77e7f769... · File a0c92a6cfdcba39b...

```
146 |         case ed25519.PrivateKey:
147 |                 return &ed25519PrivateKey{&k}, nil
148 |         case *rsa.PrivateKey:
```

file_excerpt · bccsp/sw/fileks.go · line 164 · SHA-256 749ebaa93ccb6808... · File a0c92a6cfdcba39b...

```
163 |         case ed25519.PublicKey:
164 |                 return &ed25519PublicKey{&k}, nil
165 |         case *rsa.PublicKey:
```

file_excerpt · bccsp/sw/fileks.go · line 198 · SHA-256 070f2f8ff0382634... · File a0c92a6cfdcba39b...

```
197 |
198 | case *ed25519PrivateKey:
199 |         err = ks.storePrivateKey(hex.EncodeToString(k.SKI()), kk.privKey)
```

file_excerpt · bccsp/sw/keygen.go · line 36 · SHA-256 07b0787395320c0d... · File 51177e8239b008e1...

```
35 | func (kg *ed25519KeyGenerator) KeyGen(opts bccsp.KeyGenOpts) (bccsp.Key, error) {
36 | _, privKey, err := ed25519.GenerateKey(rand.Reader)
37 | if err != nil {
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17513 · CODE-QV-ASYMMETRIC

Medium

Quantum-vulnerable RSA: role not determined (6 file(s))

Long-term post-quantum risk | Confidence: Medium | preliminary — requires further verification

7 reference(s) to RSA across 6 file(s), e.g. bccsp/sw/fileks.go:13. Role, judged from paths, function names and nearby code: role not determined. The code does not show what the keys protect; review decides the priority.

Affected asset	source-code:RSA:general
Data source	bccsp/sw/fileks.go
Detection method	Regular-expression matching of crypto API patterns; the role comes from words in paths, enclosing function names and the lines around the matches.
Analyzer	source_code 1.3.0 · 2026-11-06 10:15 UTC
Priority score	34.2 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable
Mosca's inequality (x + y > z)	x=10.0, y=5.0, z=8.0 → migration would finish too late under this scenario

POTENTIAL IMPACT

Long-term risk depends on what the algorithm protects: signatures (authenticity until migration) or key exchange/encryption (retroactive decryption of recorded data).

REPRODUCTION STEPS

```
Open bccsp/sw/fileks.go:13 and inspect the quoted lines; the evidence lists further files.
```

EVIDENCE

file_excerpt · bccsp/sw/fileks.go · line 13 · SHA-256 768ee06b9c3e2ed1… · File a0c92a6cfdcba39b…

```
12 | "crypto/ed25519"
13 | "crypto/rsa"
14 | "encoding/hex"
```

file_excerpt · bccsp/sw/keygen.go · line 14 · SHA-256 364e7ffecee10ae3… · File 51177e8239b008e1…

```
13 | "crypto/rand"
14 | "crypto/rsa"
15 | "fmt"
```

file_excerpt · bccsp/sw/keygen.go · line 62 · SHA-256 b7bba74d01f03eef… · File 51177e8239b008e1…

```
61 | func (kg *rsaKeyGenerator) KeyGen(opts bccsp.KeyGenOpts) (bccsp.Key, error) {
62 | lowLevelKey, err := rsa.GenerateKey(rand.Reader, int(kg.length))
63 | if err != nil {
```

file_excerpt · bccsp/sw/keyimport.go · line 12 · SHA-256 7318348444f9d7d1… · File 573e238eea23ae85…

```
11 | "crypto/ed25519"
12 | "crypto/rsa"
13 | "crypto/x509"
```

file_excerpt · bccsp/sw/keys.go · line 15 · SHA-256 070992b7bb94b0db… · File cd0a974ae8900553…

```
14 | "crypto/rand"
15 | "crypto/rsa"
16 | "crypto/x509"
```

file_excerpt · bccsp/sw/rsa.go · line 11 · SHA-256 a77c4604bea38eaa… · File e3112f8c38642c19…

```
10 | "crypto/rand"
11 | "crypto/rsa"
12 | "errors"
```

file_excerpt · bccsp/sw/rsakey.go · line 10 · SHA-256 5e8f30d23b521293… · File 93838157dd6a4eca…

```
 9 | import (
10 | "crypto/rsa"
11 | "crypto/sha256"
```

LIMITATIONS OF THE CHECK

This is a forward-looking risk that depends on the arrival of a cryptographically relevant quantum computer (CRQC). No such computer is publicly known to exist today; the timeline is uncertain. It is not a present-day exploitable vulnerability. Heuristic pattern match on source text without data-flow analysis. The usage context (security-relevant or not) must be confirmed manually; treat as preliminary.

RECOMMENDATION

Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility.

RETEST PROCEDURE

Re-run the inventory after refactoring.

P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997); NIST IR 8547 — Transition to Post-Quantum Cryptography Standards; M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018)

QSI-2026-17515 · DEP-ADVISORY-UNRATED

Info

golang.org/x/net v0.58.0 is affected by GO-2026-6603 (CVE-2026-78659) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6603 (CVE-2026-78659) for Go package golang.org/x/net at version v0.58.0: HTTP/2 server memory exhaustion due to Trailer headers in net/http. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. None of the affected functions (golang.org/x/net/http2.ClientConn.Close, golang.org/x/net/http2.ClientConn.Ping, golang.org/x/net/http2.ClientConn.RoundTrip, golang.org/x/net/http2.ClientConn.Shutdown, golang.org/x/net/http2.ConfigureServer, golang.org/x/net/http2.ConfigureTransport …) is named in the audited code: the advisory does not apply unless they are reached indirectly (traits, macros, other dependencies).

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 10:15 UTC

Priority score	17.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6603
```

EVIDENCE

tool_output · OSV.dev GO-2026-6603 · line 46 · SHA-256 9e43850e0da48e9d…


```
{"id": "GO-2026-6603", "aliases": ["BIT-golang-2026-78659", "CVE-2026-78659"], "modified":
"2026-10-16T09:55:47.068117441Z", "summary": "HTTP/2 server memory exhaustion due to Trailer headers
in net/http", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed. A Go function counts only in a file that also names its package (a method: its type); unexported functions only through the exported ones.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.60.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6603

QSI-2026-17516 · DEP-ADVISORY-UNRATED

Info

golang.org/x/net v0.58.0 is affected by GO-2026-6610 (CVE-2026-78660) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6610 (CVE-2026-78660) for Go package golang.org/x/net at version v0.58.0: HTTP/2 transport accepts malformed framing-related headers in net/http. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. None of the affected functions (golang.org/x/net/http2.Transport.NewClientConn, golang.org/x/net/http2.Transport.RoundTrip, golang.org/x/net/http2.Transport.RoundTripOpt, golang.org/x/net/http2.clientConnPool.GetClientConn, golang.org/x/net/http2.clientConnReadLoop.handleResponse, golang.org/x/net/http2.noDialClientConnPool.GetClientConn …) is named in the audited code: the advisory does not apply unless they are reached indirectly (traits, macros, other dependencies).

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.

Analyzer	dependencies 1.3.0 · 2026-11-06 10:15 UTC
Priority score	17.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6610
```

EVIDENCE

tool_output · OSV.dev GO-2026-6610 · line 46 · SHA-256 1b823e53f2709379...


```
{"id": "GO-2026-6610", "aliases": ["BIT-golang-2026-78660", "CVE-2026-78660"], "modified":
"2026-10-16T09:56:04.364467528Z", "summary": "HTTP/2 transport accepts malformed framing-related
headers in net/http", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed. A Go function counts only in a file that also names its package (a method: its type); unexported functions only through the exported ones.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.60.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6610

QSI-2026-17517 · DEP-ADVISORY-UNRATED

Info

golang.org/x/net v0.58.0 is affected by GO-2026-6611 (CVE-2026-78669) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6611 (CVE-2026-78669) for Go package golang.org/x/net at version v0.58.0: Excessive CPU consumption from repeated initial window changes in net/http. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. None of the affected functions (golang.org/x/net/http2.ClientConn.Close, golang.org/x/net/http2.ClientConn.Ping, golang.org/x/net/http2.ClientConn.RoundTrip, golang.org/x/net/http2.ClientConn.Shutdown, golang.org/x/net/http2.ClientConn.addStreamLocked, golang.org/x/net/http2.ConfigureServer ...) is named in the audited code: the advisory does not apply unless they are reached indirectly (traits, macros, other dependencies).

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.

Analyzer	dependencies 1.3.0 · 2026-11-06 10:15 UTC
Priority score	17.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6611
```

EVIDENCE

tool_output · OSV.dev GO-2026-6611 · line 46 · SHA-256 c9b9f78637131224…


```
{"id": "GO-2026-6611", "aliases": ["BIT-golang-2026-78669", "CVE-2026-78669"], "modified":
"2026-10-16T09:55:38.952277866Z", "summary": "Excessive CPU consumption from repeated initial window
changes in net/http", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed. A Go function counts only in a file that also names its package (a method: its type); unexported functions only through the exported ones.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.60.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6611

QSI-2026-17518 · DEP-ADVISORY-UNRATED

Info

golang.org/x/net v0.58.0 is affected by GO-2026-6612 (CVE-2026-78663) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6612 (CVE-2026-78663) for Go package golang.org/x/net at version v0.58.0: Double flow control refund on HTTP/2 server streams in net/http. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. None of the affected functions (golang.org/x/net/http2.ClientConn.Close, golang.org/x/net/http2.ClientConn.Ping, golang.org/x/net/http2.ClientConn.RoundTrip, golang.org/x/net/http2.ClientConn.Shutdown, golang.org/x/net/http2.ConfigureServer, golang.org/x/net/http2.ConfigureTransport …) is named in the audited code: the advisory does not apply unless they are reached indirectly (traits, macros, other dependencies).

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.

Analyzer	dependencies 1.3.0 · 2026-11-06 10:15 UTC
Priority score	17.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6612
```

EVIDENCE

tool_output · OSV.dev GO-2026-6612 · line 46 · SHA-256 1471910f88c9bbf8...


```
{"id": "GO-2026-6612", "aliases": ["BIT-golang-2026-78663", "CVE-2026-78663"], "modified": "2026-10-16T09:55:54.129082194Z", "summary": "Double flow control refund on HTTP/2 server streams in net/http", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed. A Go function counts only in a file that also names its package (a method: its type); unexported functions only through the exported ones.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.60.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6612

QSI-2026-17519 · DEP-ADVISORY-UNRATED

Info

golang.org/x/net v0.58.0 is affected by GO-2026-6617 (CVE-2026-97032) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6617 (CVE-2026-97032) for Go package golang.org/x/net at version v0.58.0: HTTP/2 server crash due to HPACK encoder race in net/http. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. None of the affected functions (golang.org/x/net/http2.ClientConn.Close, golang.org/x/net/http2.ClientConn.Ping, golang.org/x/net/http2.ClientConn.RoundTrip, golang.org/x/net/http2.ClientConn.Shutdown, golang.org/x/net/http2.ConfigureServer, golang.org/x/net/http2.ConfigureTransport ...) is named in the audited code: the advisory does not apply unless they are reached indirectly (traits, macros, other dependencies).

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.

Analyzer	dependencies 1.3.0 · 2026-11-06 10:15 UTC
Priority score	17.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6617
```

EVIDENCE

tool_output · OSV.dev GO-2026-6617 · line 46 · SHA-256 b948c88d959bc789…


```
{"id": "GO-2026-6617", "aliases": ["BIT-golang-2026-97032", "CVE-2026-97032"], "modified":
"2026-10-16T09:55:57.350751224Z", "summary": "HTTP/2 server crash due to HPACK encoder race in net/
http", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed. A Go function counts only in a file that also names its package (a method: its type); unexported functions only through the exported ones.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.60.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6617

QSI-2026-17520 · DEP-ADVISORY-UNRATED

Info

golang.org/x/net v0.58.0 is affected by GO-2026-6860 (CVE-2026-78661) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6860 (CVE-2026-78661) for Go package golang.org/x/net at version v0.58.0: Avoid quadratic complexity when coalescing text nodes in html in golang.org/x/net. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. None of the affected functions (golang.org/x/net/html.Parse, golang.org/x/net/html.ParseFragment, golang.org/x/net/html.ParseFragmentWithOptions, golang.org/x/net/html.ParseWithOptions, golang.org/x/net/html.parser.addText, golang.org/x/net/html.parser.fosterParent …) is named in the audited code: the advisory does not apply unless they are reached indirectly (traits, macros, other dependencies).

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.

Analyzer	dependencies 1.3.0 · 2026-11-06 10:15 UTC
Priority score	17.8 / 100 (Crypto exchange, custodian or wallet)
CVSS v3.1	not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6860
```

EVIDENCE

tool_output · OSV.dev GO-2026-6860 · line 46 · SHA-256 1f106c45371219ba...


```
{"id": "GO-2026-6860", "aliases": ["CVE-2026-78661"], "modified": "2026-10-27T21:30:04.890809062Z",
"summary": "Avoid quadratic complexity when coalescing text nodes in html in golang.org/x/net",
"severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed. A Go function counts only in a file that also names its package (a method: its type); unexported functions only through the exported ones.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.62.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6860

QSI-2026-17521 · DEP-ADVISORY-UNRATED

Info

golang.org/x/net v0.58.0 is affected by GO-2026-6866 (CVE-2026-97045) (no severity rating)

Known vulnerable component | Confidence: Low | preliminary — requires further verification

OSV.dev reports GO-2026-6866 (CVE-2026-97045) for Go package golang.org/x/net at version v0.58.0: Check all ancestor namespace transitions when rendering raw text in html in golang.org/x/net. The advisory carries neither a CVSS vector nor a database severity, so QSI rates it low and preliminary until it is triaged. None of the affected functions (golang.org/x/net/html.Render, golang.org/x/net/html.childTextNodesAreLiteral) is named in the audited code: the advisory does not apply unless they are reached indirectly (traits, macros, other dependencies).

Affected asset	dependency:Go/golang.org/x/net@v0.58.0
Data source	OSV.dev
Detection method	OSV.dev querybatch API lookup of the exact declared version.
Analyzer	dependencies 1.3.0 · 2026-11-06 10:15 UTC
Priority score	17.8 / 100 (Crypto exchange, custodian or wallet)

CVSS v3.1 not applicable

POTENTIAL IMPACT

Unknown without triage: read the advisory and check whether the affected code is reachable in your build.

REPRODUCTION STEPS

```
curl -s https://api.osv.dev/v1/vulns/GO-2026-6866
```

EVIDENCE

tool_output · OSV.dev GO-2026-6866 · line 46 · SHA-256 102267a016bc5eef…


```
{"id": "GO-2026-6866", "aliases": ["CVE-2026-97045"], "modified": "2026-10-27T21:45:08.978077289Z",
"summary": "Check all ancestor namespace transitions when rendering raw text in html in golang.org/
x/net", "severity": null, "declared_in_manifest": true}
```


LIMITATIONS OF THE CHECK

Advisory data comes from OSV.dev at the recorded time. The advisory has no severity rating. Reachability is judged from names, not a call graph: the affected functions the advisory lists against the identifiers in the audited product code, a Cargo feature it says is off by default against the audited manifests, the platforms it lists against the declared Rust targets. Indirect calls (traits, macros, other dependencies) are not followed. A Go function counts only in a file that also names its package (a method: its type); unexported functions only through the exported ones.

RECOMMENDATION

Triage the advisory; upgrade to a fixed version (0.62.0) if it applies.

RETEST PROCEDURE

Re-run the dependency scan after upgrading.

OSV advisory GO-2026-6866

Evidence

ID	Type	Data source	SHA-256
QSI-2026-17514	tool_output	OSV.dev GHSA-2v4p-qf9q-27wj (line 19)	3566bf539cd1bed726145fc02332a18308f9a065bb8b992c0e44deb9ec44221a
QSI-2026-17511	file_excerpt	bccsp/pkcs11/ecdsakey.go (line 19)	87680e3cf321a1d927f748c94e56fef66319eb13e2554469b4d613e55886927b
QSI-2026-17511	file_excerpt	bccsp/pkcs11/pkcs11.go (line 11)	8da8b0c88ebe01946640787c08a65d84eb065c628bc69992d0bd8b7d1eda1ff8
QSI-2026-17511	file_excerpt	bccsp/pkcs11/pkcs11.go (line 330)	c46c43f1ab599be0dc6777bac06d578b05466bf53f5527162b89cacfcc3b671c
QSI-2026-17511	file_excerpt	bccsp/sw/ecdsa.go (line 19)	1515eca1cc78dc2b7c4fde09c0aa059dc7c5a6a53519f5c1b42098f4eadc395d
QSI-2026-17511	file_excerpt	bccsp/sw/ecdsa.go (line 28)	5e03fc2ced291506b09bf90d810db18d327d592f8781b028b97d572b077c587c
QSI-2026-17511	file_excerpt	bccsp/sw/ecdsa.go (line 56)	cc80b9b87564d0ae70ba183543c926e98984365178ee8a10c6384959a648b188
QSI-2026-17511	file_excerpt	bccsp/sw/ecdsakey.go (line 19)	da30811132376f3039c613505f0d52458e1593d43e0280ad13aaf57389cf7c73
QSI-2026-17511	file_excerpt	bccsp/sw/fileks.go (line 11)	e5b44015f00ef128ab2f3d1c4f1dee7dcd77a13dd2b3391fad9bc2ecb9c49ae6
QSI-2026-17511	file_excerpt	bccsp/sw/keyderiv.go (line 20)	9f85b73ac6ec142259370ca9358f25e99f4a2d13e133f7ddee639c918ed9af9b
QSI-2026-17511	file_excerpt	bccsp/sw/keygen.go (line 10)	7983932d009be78f478f371ae58c3f86b9f6b72f687b9ed011a4aee4ad95d71d
QSI-2026-17512	file_excerpt	bccsp/sw/ed25519.go (line 15)	be962610d24d0e25089b18a267e417de6efea2a91e686d6803d7a04d54b8cb0c
QSI-2026-17512	file_excerpt	bccsp/sw/ed25519.go (line 20)	aa94dd71ac249c8771fc540b4bc0ae1102dafe73a0a1f2722a94d7ba614d5d79

QSI-2026-17512	file_excerpt	bccsp/sw/ed25519.go (line 23)	`6f33081caebdef556a3b8c3fe6eeca1f1c206bd80023c04d0dd16a68a6634e70`
QSI-2026-17512	file_excerpt	bccsp/sw/ed25519key.go (line 18)	`c161f2e6d9d89ebe1d96be52c121110e7611fbcf8cb5872b5816d483a91b9152`
QSI-2026-17512	file_excerpt	bccsp/sw/ed25519key.go (line 24)	`ee921a0dccc1f590e41024cdb8f552d96e7e79860fc51361bdd9e7282a296bd9`
QSI-2026-17512	file_excerpt	bccsp/sw/ed25519key.go (line 29)	`488f2f75c771d7f192ac75a4d1524cd88cad84dc34abaa8f25806c126cb56d58`
QSI-2026-17512	file_excerpt	bccsp/sw/fileks.go (line 147)	`1cfe306c77e7f7694c28b313f33d969f772a0285e232590afa48e1c614905b2c`
QSI-2026-17512	file_excerpt	bccsp/sw/fileks.go (line 164)	`749ebaa93ccb680875e11da118e180004d476946a6c4e4bcacbc5ac6dc3056b6`
QSI-2026-17512	file_excerpt	bccsp/sw/fileks.go (line 198)	`070f2f8ff0382634eed998627dc8b68a8775020445d49702f7472219ea427e14`
QSI-2026-17512	file_excerpt	bccsp/sw/keygen.go (line 36)	`07b0787395320c0dcb325ee82d4a64a304cd8fc2872374aa7dd3cfc78e7191c7`
QSI-2026-17513	file_excerpt	bccsp/sw/fileks.go (line 13)	`768ee06b9c3e2ed1d6af2ed858f328bdd5acdddaccbe6d3c2ecd30aebb1ab1d7`
QSI-2026-17513	file_excerpt	bccsp/sw/keygen.go (line 14)	`364e7ffecee10ae3b8e624ca22f247d6c912d5e4981f71e890c23ab5677cf5dd`
QSI-2026-17513	file_excerpt	bccsp/sw/keygen.go (line 62)	`b7bba74d01f03eef780ca8edc8d8f543670a53cc919dc442239153bb3820145c`
QSI-2026-17513	file_excerpt	bccsp/sw/keyimport.go (line 12)	`7318348444f9d7d18c85c7efbf08d2965e7d68412a40ee459b218cde70685d97`
QSI-2026-17513	file_excerpt	bccsp/sw/keys.go (line 15)	`070992b7bb94b0dbc8e66fb1b57b459fe619fa742bac22418518a3f923a61864`
QSI-2026-17513	file_excerpt	bccsp/sw/rsa.go (line 11)	`a77c4604bea38eaadc0ee9cf5c192333b6ac97a6a6ab944dce48ee75a086b6a3`
QSI-2026-17513	file_excerpt	bccsp/sw/rsakey.go (line 10)	`5e8f30d23b521293b331ecad3ad4d4d3b8072be834d89f4c17cd5a1a5cad919a`

QSI-2026-17515	tool_output	OSV.dev GO-2026-6603 (line 46)	`9e43850e0da48e9d00589cc2bffb372f39ec5e5a280f2f4b04520a91328fd703`
QSI-2026-17516	tool_output	OSV.dev GO-2026-6610 (line 46)	`1b823e53f27093797b8a7909032006e1f0f818f04118ad593a4641236392faa9`
QSI-2026-17517	tool_output	OSV.dev GO-2026-6611 (line 46)	`c9b9f78637131224c5d4073e44c47b7cce490fc5526e3d6b0697b43bc85cab2f`
QSI-2026-17518	tool_output	OSV.dev GO-2026-6612 (line 46)	`1471910f88c9bbf8395a304fd82510f61567373756002de8bcdc3fbe305890cb`
QSI-2026-17519	tool_output	OSV.dev GO-2026-6617 (line 46)	`b948c88d959bc78948c7e3b7a746ac4da179933798e9735d02870080906cfc96`
QSI-2026-17520	tool_output	OSV.dev GO-2026-6860 (line 46)	`1f106c45371219bac0158f7ce44e57c439f50f0c32898616d1563f14cadb1b28`
QSI-2026-17521	tool_output	OSV.dev GO-2026-6866 (line 46)	`102267a016bc5eef7979a90b1ddd2288ff36b09ab6fa040c808e62f3b55c1f42`

Post-quantum assessment

	Key establishment (x = how long data must stay confidential; harvest now, decrypt later)	Signatures (x = how long signatures/ keys must stay trusted)
Data/key lifetime x	10	10
Migration time y	5	5
Assumed years to CRQC z	8 (early)	8
x + y − z	7 → migration would finish too late under this scenario	7 → migration would finish too late under this scenario

Planning assumptions: x for key establishment, x for signatures, y are the defaults of the risk profile “Crypto exchange, custodian or wallet”, not the client's figures; with the client's actual lifetimes the verdict can change.

Post-quantum maturity: Level 0 of 4 — no post-quantum cryptography

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

Levels: 0 none; 1 present, not wired; 2 in use outside the critical paths; 3 hybrid in a critical path; 4 post-quantum by default in the critical paths (root of trust, key wrapping, channel key exchange). Judged from code and manifests.

What the quantum-vulnerable algorithms do

Role	Algorithms (files)	ID	First step
role not determined	ECDSA (10), Ed25519 (6), RSA (6)	QSI-2026-17511, QSI-2026-17512, QSI-2026-17513	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it; introduce an abstraction layer for crypto-agility

Roles are judged from file paths, function names and the code around each match; review confirms them. Root-of-trust keys and wrapped keys come first: they are hard to replace or can be recorded today.

z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Quantum status	Count
reduced margin (Grover), approved	4
Grover: acceptable to NIST, below ECCG/ANSSI guidance	9
vulnerable (Shor)	24

Detailed findings: QSI-2026-17511, QSI-2026-17512, QSI-2026-17513

PQC readiness questionnaire

No readiness questionnaire was completed for this project.

Quantum algorithms relevant to this inventory

Quantum algorithm	Effect	Inventoried algorithms (rows)	What it means for this audit
Shor's algorithm — factoring Q1: quantum computation offline, classical access to the victim	breaks	RSA (6)	Breaks RSA at every key length: a longer modulus only raises the cost polynomially. Plan migration to ML-KEM (or a hybrid) for key establishment and ML-DSA / SLH-DSA for signatures; start with keys protecting data that must stay confidential for years (harvest now, decrypt later) and with long-lived signatures.
Shor's algorithm — elliptic-curve discrete logarithms Q1: quantum computation offline, classical access to the victim	breaks	ECDSA (10), Ed25519 (8)	Elliptic-curve keys are expected to fall before RSA keys of comparable classical strength. On blockchains, coins whose public key is already on chain (reused addresses, pay-to-public-key and Taproot outputs) are exposed first; plan the move to post-quantum signatures and avoid key reuse in the meantime.
Grover's algorithm — key search and preimages Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	HMAC (2), SHA-2 (9), SHA-3 (2)	Not a break. 256-bit keys (AES-256, ChaCha20) keep a comfortable margin; AES-128 meets NIST category 1, while some authorities ask for 256-bit keys in quantum-resistant contexts (see the AES-128 entry). Prefer 256-bit keys for long-lived data; 256-bit hashes need no change for preimage resistance.
Brassard–Høyer–Tapp — collision search Q1: quantum computation offline, classical access to the victim	weakens (larger keys restore the margin)	SHA-2 (9), SHA-3 (2)	256-bit hashes (SHA-256, SHA3-256) remain adequate for collision resistance; 160-bit hashes are already weak classically (SHA-1 collisions are practical). Use 384/512-bit hashes where NIST categories 4–5 are required.

Effects: breaks — replace the algorithm; weakens — larger keys or outputs restore the margin; included in standard parameters — no action at standard parameter sets; only with superposition queries — not applicable to ordinary deployments.

Limitations and assumptions

- This report presents automated and documented analysis results for the stated scope. It is not a legal opinion, a certification or an attestation of compliance with any standard.
- z is a planning assumption chosen per risk profile, not a prediction. A cryptographically relevant quantum computer is not publicly known to exist today.

Remediation plan

QSI-2026-17514 · P2 · 30 days root-cause fix

Upgrade google.golang.org/grpc to a fixed version (1.82.2, 1.83.2, 1.84.0-dev.0.20260825144003-d5a41119e0e3, 1.85.0-dev.0.20260825072537-93e31b48545e).

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GHSA-2v4p-qf9q-27wj (CVE-2026-84445, GO-2026-6443).

QSI-2026-17515 · P3 · 60 days root-cause fix

Triage GO-2026-6603 (CVE-2026-78659); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6603 (CVE-2026-78659), or the triage is recorded.

QSI-2026-17516 · P3 · 60 days root-cause fix

Triage GO-2026-6610 (CVE-2026-78660); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6610 (CVE-2026-78660), or the triage is recorded.

QSI-2026-17517 · P3 · 60 days root-cause fix

Triage GO-2026-6611 (CVE-2026-78669); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6611 (CVE-2026-78669), or the triage is recorded.

QSI-2026-17518 · P3 · 60 days root-cause fix

Triage GO-2026-6612 (CVE-2026-78663); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6612 (CVE-2026-78663), or the triage is recorded.

QSI-2026-17519 · P3 · 60 days root-cause fix

Triage GO-2026-6617 (CVE-2026-97032); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6617 (CVE-2026-97032), or the triage is recorded.

QSI-2026-17520 · P3 · 60 days root-cause fix

Triage GO-2026-6860 (CVE-2026-78661); upgrade golang.org/x/net to a fixed version (0.62.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6860 (CVE-2026-78661), or the triage is recorded.

QSI-2026-17521 · P3 · 60 days root-cause fix

Triage GO-2026-6866 (CVE-2026-97045); upgrade golang.org/x/net to a fixed version (0.62.0) if it applies.

Rationale	Eliminates the root cause described in the finding.
Expected outcome	The weakness is no longer present on re-test.
Complexity	low
Dependencies	Change window and owner approval.
Side effects	Possible API changes.
Tests before change	Reproduce the finding using the documented steps.
Tests after change	Re-run the retest procedure from the finding.
Closure criteria	Dependency scan no longer reports GO-2026-6866 (CVE-2026-97045), or the triage is recorded.

QSI-2026-17511 · P3 · 90 days short-term risk reduction

Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17512 · P3 · 90 days

short-term risk reduction

Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17513 · P3 · 90 days

short-term risk reduction

Add RSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).

Rationale	Migration cannot be planned without knowing where and why the algorithm is used.
Expected outcome	Every usage has an owner, a lifetime and a migration path.
Complexity	medium
Dependencies	Access to system owners.
Side effects	None (documentation and design work).
Tests before change	Confirm the inventory entry matches the deployed configuration.
Tests after change	Inventory review signed off by the system owner.
Closure criteria	Inventory entry with owner, lifetime and target algorithm exists.

QSI-2026-17511 · P4 · 90 days

root-cause fix

Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-17512 · P4 · 90 days root-cause fix

Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

QSI-2026-17513 · P4 · 90 days root-cause fix

Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.

Rationale	Removes dependence on a quantum-vulnerable public-key primitive.
Expected outcome	The asset uses a NIST-standardized post-quantum or hybrid algorithm.
Complexity	high
Dependencies	Library/protocol support (e.g. OpenSSL ≥ 3.5, HSM firmware), interoperability with peers.
Side effects	Larger keys/signatures, possible performance and compatibility impact.
Tests before change	Interoperability tests in a staging environment.
Tests after change	Verify the negotiated/used algorithm on the production asset.
Closure criteria	Re-audit shows ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/ composite form where the ecosystem supports it or a hybrid scheme in use.

Priorities for 30, 60 and 90 days

30 days

Priority	ID	Action	Type
P2	QSI-2026-17514	Upgrade google.golang.org/grpc to a fixed version (1.82.2, 1.83.2, 1.84.0-dev.0.20260825144003-d5a41119e0e3, 1.85.0-dev.0.20260825072537-93e31b48545e).	root-cause fix

60 days

Priority	ID	Action	Type
P3	QSI-2026-17515	Triage GO-2026-6603 (CVE-2026-78659); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.	root-cause fix
P3	QSI-2026-17516	Triage GO-2026-6610 (CVE-2026-78660); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.	root-cause fix
P3	QSI-2026-17517	Triage GO-2026-6611 (CVE-2026-78669); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.	root-cause fix
P3	QSI-2026-17518	Triage GO-2026-6612 (CVE-2026-78663); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.	root-cause fix
P3	QSI-2026-17519	Triage GO-2026-6617 (CVE-2026-97032); upgrade golang.org/x/net to a fixed version (0.60.0) if it applies.	root-cause fix
P3	QSI-2026-17520	Triage GO-2026-6860 (CVE-2026-78661); upgrade golang.org/x/net to a fixed version (0.62.0) if it applies.	root-cause fix
P3	QSI-2026-17521	Triage GO-2026-6866 (CVE-2026-97045); upgrade golang.org/x/net to a fixed version (0.62.0) if it applies.	root-cause fix

90 days

Priority	ID	Action	Type
P3	QSI-2026-17511	Add ECDSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17512	Add Ed25519 usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P3	QSI-2026-17513	Add RSA usages at source code to the cryptographic inventory with an owner and a data-lifetime estimate; design for crypto-agility (algorithm identifiers, configurable providers).	short-term risk reduction
P4	QSI-2026-17511	Plan migration of ECDSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

P4	QSI-2026-17512	Plan migration of Ed25519 to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix
P4	QSI-2026-17513	Plan migration of RSA to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form where the ecosystem supports it (or a hybrid scheme) once supported by your libraries, protocols and counterparties.	root-cause fix

Verification after remediation

ID	Retest procedure
QSI-2026-17514	Re-run the dependency scan after upgrading.
QSI-2026-17511	Re-run the inventory after refactoring.
QSI-2026-17512	Re-run the inventory after refactoring.
QSI-2026-17513	Re-run the inventory after refactoring.
QSI-2026-17515	Re-run the dependency scan after upgrading.
QSI-2026-17516	Re-run the dependency scan after upgrading.
QSI-2026-17517	Re-run the dependency scan after upgrading.
QSI-2026-17518	Re-run the dependency scan after upgrading.
QSI-2026-17519	Re-run the dependency scan after upgrading.
QSI-2026-17520	Re-run the dependency scan after upgrading.
QSI-2026-17521	Re-run the dependency scan after upgrading.

Tools used

Tool	Version
blockchain	1.0.0
certificates	1.2.0
chain_map	1.0.0
configs	1.1.0
dependencies	1.3.0
inventory_document	1.1.0
solidity	1.3.0
source_code	1.4.0
tls_code	1.1.0
tls_live	1.0.0
qsi-engine	1.3.0
OSV.dev API	v1

Sources and normative documents

Reviewed: 2026-10-13

1. A. Chailloux, M. Naya-Plasencia, A. Schrottenloher — An Efficient Quantum Collision Search Algorithm and Implications on Symmetric Cryptography (ASIACRYPT 2017; IACR ePrint 2017/847) — IACR ePrint. https://eprint.iacr.org/2017/847 (peer-reviewed article: Õ(2^{2n/5}) time, O(n) qubits, Õ(2^{n/5}) classical memory)
2. C. Chevignard, P.-A. Fouque, A. Schrottenloher — Reducing the Number of Qubits in Quantum Discrete Logarithms on Elliptic Curves (IACR ePrint 2026/280) — IACR ePrint. https://eprint.iacr.org/2026/280 (revision of a EUROCRYPT 2026 paper: 3.12n + o(n) qubits; for n = 256, 1193 qubits and 22 runs of 2^38.98 Toffoli gates)
3. C. Gidney — How to factor 2048 bit RSA integers with less than a million noisy qubits (arXiv:2505.15917) — arXiv. https://arxiv.org/abs/2505.15917 (preprint (2025-05-21): < 1 million noisy qubits, < 1 week, 0.1% gate error, 1 µs surface-code cycle)
4. C. Gidney, M. Ekerå — How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits (Quantum 5, 433) — Quantum. https://doi.org/10.22331/q-2021-04-15-433 (peer-reviewed article (2021): 10^-3 gate error, 1 µs cycle, 10 µs reaction time)
5. C. Zalka — Grover's quantum searching algorithm is optimal (Phys. Rev. A 60, 2746, 1999; arXiv:quant-ph/9711070) — APS. https://doi.org/10.1103/PhysRevA.60.2746 (peer-reviewed article: optimal for any success probability; parallel search gains no more than splitting the space)
6. Common Vulnerability Scoring System v3.1: Specification Document — FIRST. https://www.first.org/cvss/v3.1/specification-document (final)
7. D. J. Bernstein — Cost analysis of hash collisions: Will quantum computers make SHARCS obsolete? (2009) — cr.yp.to. https://cr.yp.to/hash/collisioncost-20090823.pdf (paper dated 2009-08-23)
8. D. Litinski — How to compute a 256-bit elliptic curve private key with only 50 million Toffoli gates (arXiv:2306.08585) — arXiv. https://arxiv.org/abs/2306.08585 (preprint (2023-06-14): 6000 modules × 1152 physical qubits, one key per 10 minutes under its architecture assumptions)
9. FIPS 203 — Module-Lattice-Based Key-Encapsulation Mechanism Standard (ML-KEM) — NIST. https://csrc.nist.gov/pubs/fips/203/final (final (August 2024))
10. FIPS 204 — Module-Lattice-Based Digital Signature Standard (ML-DSA) — NIST. https://csrc.nist.gov/pubs/fips/204/final (final (August 2024))
11. FIPS 205 — Stateless Hash-Based Digital Signature Standard (SLH-DSA) — NIST. https://csrc.nist.gov/pubs/fips/205/final (final (August 2024))
12. G. Brassard, P. Høyer, A. Tapp — Quantum Algorithm for the Collision Problem (LATIN '98; arXiv:quant-ph/9705002) — Springer / arXiv. https://arxiv.org/abs/quant-ph/9705002 (peer-reviewed article: collisions of r-to-one functions in O((N/r)^{1/3}) expected evaluations)
13. J. Proos, C. Zalka — Shor's discrete logarithm quantum algorithm for elliptic curves (Quantum Inf. Comput. 3(4), 2003; arXiv:quant-ph/0301141) — Quantum Information & Computation. https://doi.org/10.26421/QIC3.4-3 (peer-reviewed article)
14. L. K. Grover — A fast quantum mechanical algorithm for database search (STOC 1996) — ACM. https://doi.org/10.1145/237814.237866 (peer-reviewed article)
15. M. Amy et al. — Estimating the cost of generic quantum pre-image attacks on SHA-2 and SHA-3 (SAC 2016; arXiv:1603.09383) — Springer / arXiv. https://arxiv.org/abs/1603.09383 (peer-reviewed article: SHA-256 about 2^153.8 surface-code cycles on 2^12.6 logical qubits; SHA3-256 about 2^146.5 cycles on 2^20)
16. M. Cain et al. — Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits (arXiv:2603.28627) — arXiv. https://arxiv.org/abs/2603.28627 (preprint (2026-03-30): P-256 in a few days with 26,000 physical neutral-atom qubits; RSA-2048 one to two orders of magnitude longer)
17. M. Mosca — Cybersecurity in an Era with Quantum Computers: Will We Be Ready? (IEEE Security & Privacy, 2018) — IEEE. https://doi.org/10.1109/MSP.2018.3761723 (peer-reviewed article; source of the x + y > z migration inequality)
18. M. Roetteler, M. Naehrig, K. Svore, K. Lauter — Quantum resource estimates for computing elliptic curve discrete logarithms (ASIACRYPT 2017) — IACR / arXiv. https://arxiv.org/abs/1706.06752 (peer-reviewed article: at most 9n + 2⌈log2 n⌉ + 10 qubits and 448 n^3 log2 n + 4090 n^3 Toffoli gates)
19. NIST IR 8547 — Transition to Post-Quantum Cryptography Standards — NIST. https://csrc.nist.gov/pubs/ir/8547/final (final (2026-10-27; the initial public draft of November 2024 is obsoleted). Tables 2 and 4: ECDSA, EdDSA and RSA signatures, finite-field and elliptic-curve DH and MQV, and RSA key establishment are deprecated after 2030 and disallowed after 2035 at every key size — the draft deprecated only the 112-bit level. A composite (hybrid) key establishment stays FIPS 140-validatable while one input comes from an approved scheme. Repeats the federal deadlines of OMB M-26-15: post-quantum key establishment by 2030-12-31 and signatures by 2031-12-31 for high value assets and high impact systems)
20. NIST — Post-Quantum Cryptography Standardization: Evaluation Criteria, Security (Call for Proposals, Section 4.A.5) — NIST. https://csrc.nist.gov/Projects/Post-Quantum-Cryptography/Post-Quantum-Cryptography-Standardization/

Evaluation-Criteria/Security-(Evaluation-Criteria) (security categories: AES-128 key search 2^170/MAXDEPTH quantum gates, AES-192 2^233/MAXDEPTH, AES-256 2^298/MAXDEPTH; MAXDEPTH 2^40–2^96)
21. OSV advisory GHSA-2v4p-qf9q-27wj — OSV.dev. https://osv.dev/vulnerability/GHSA-2v4p-qf9q-27wj (modified 2026-09-25T18:00:05.270054427Z)
22. OSV advisory GO-2026-6603 — OSV.dev. https://osv.dev/vulnerability/GO-2026-6603 (modified 2026-10-16T09:55:47.068117441Z)
23. OSV advisory GO-2026-6610 — OSV.dev. https://osv.dev/vulnerability/GO-2026-6610 (modified 2026-10-16T09:56:04.364467528Z)
24. OSV advisory GO-2026-6611 — OSV.dev. https://osv.dev/vulnerability/GO-2026-6611 (modified 2026-10-16T09:55:38.952277866Z)
25. OSV advisory GO-2026-6612 — OSV.dev. https://osv.dev/vulnerability/GO-2026-6612 (modified 2026-10-16T09:55:54.129082194Z)
26. OSV advisory GO-2026-6617 — OSV.dev. https://osv.dev/vulnerability/GO-2026-6617 (modified 2026-10-16T09:55:57.350751224Z)
27. OSV advisory GO-2026-6860 — OSV.dev. https://osv.dev/vulnerability/GO-2026-6860 (modified 2026-10-27T21:30:04.890809062Z)
28. OSV advisory GO-2026-6866 — OSV.dev. https://osv.dev/vulnerability/GO-2026-6866 (modified 2026-10-27T21:45:08.978077289Z)
29. P. W. Shor — Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer (SIAM J. Comput., 1997) — SIAM. https://doi.org/10.1137/S0097539795293172 (peer-reviewed article)
30. R. Babbush et al. — Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities: Resource Estimates and Mitigations (arXiv:2603.28846) — arXiv. https://arxiv.org/abs/2603.28846 (preprint (2026-03-30, v2 2026-04-15): 256-bit ECDLP with < 1200 logical qubits and < 90 M Toffoli gates (or < 1450 and < 70 M); < 500,000 physical qubits at 10^-3 error, minutes)
31. S. Jaques, M. Naehrig, M. Roetteler, F. Virdia — Implementing Grover oracles for quantum key search on AES and LowMC (EUROCRYPT 2020; arXiv:1910.01700) — Springer / arXiv. https://arxiv.org/abs/1910.01700 (peer-reviewed article: lower cost estimates for the AES key searches that define the NIST categories)

Appendices

A. Cryptographic inventory

Algorithm	Parameter	Quantum status	Classical status	Count	Locations	Examples
HMAC		reduced margin (Grover), approved	acceptable	2	2	bccsp/opts.go:53; bccsp/sw/keyderiv.go:21
SHA-2	SHA-512	reduced margin (Grover), approved	acceptable	2	2	bccsp/sw/conf.go:13; bccsp/sw/new.go:13
SHA-2	SHA-256	Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	7	7	bccsp/pkcs11/pkcs11.go:13; bccsp/sw/aeskey.go:19; bccsp/sw/conf.go:11
SHA-3		Grover: acceptable to NIST, below ECCG/ANSSI guidance	acceptable	2	2	bccsp/sw/conf.go:12; bccsp/sw/new.go:12
ECDSA		vulnerable (Shor)	acceptable	10	10	bccsp/pkcs11/ecdsakey.go:19; bccsp/pkcs11/pkcs11.go:11; bccsp/sw/ecdsa.go:19
Ed25519	Ed25519	vulnerable (Shor)	acceptable	8	8	bccsp/sw/ed25519.go:15; bccsp/sw/ed25519.go:23; bccsp/sw/ed25519key.go:18
RSA		vulnerable (Shor)	acceptable	6	6	bccsp/sw/fileks.go:13; bccsp/sw/keygen.go:14; bccsp/sw/keyimport.go:12

Long lists are grouped here; the full per-file list is in the JSON report, signed with the same key.

B. JSON

qsi.report/1 · SHA-256 967ff3502b91c3f489ae4dc96bfd80d7f227c73a45038ab60ffb4c654876bd06

C. Input files

File	Size	SHA-256	Redactions
bccsp/aesopts.go	2596	`ca0493dd6054d092027e94af3901048adc01f16192de79807d30c85378b78543`	0
bccsp/bccsp.go	4212	`82df76a75ee87e4324e9c13879bc87aa471c84db6e7d689efee6174a274ff4da`	0
bccsp/ecdsaopts.go	1499	`ed667e09393298574ad06ed64fd2cb47d855adbceca018c0d23b24654679b649`	0
bccsp/factory/factory.go	2478	`a24b6d7c7ddad62bbc52aee8c7b01fb9f3a4601ef2ed943ddb84451273e08b99`	0
bccsp/factory/nopkcs11.go	1880	`24e6532515f4ff707d8b385cf73c4a16f3d279a782a85191b3a23772dbe07c40`	0

bccsp/factory/opts.go	452	504596b431861eb308f783f1a6cb026a81f45f8f0e712411e2a9eb08ccc6cafa	0
bccsp/factory/pkcs11.go	2645	5264fbdf04137dfb54051eef9278ee7df1692e8a3c5da4c4b54b8f2f2f14f5ac	0
bccsp/factory/pkcs11factory.go	1453	c3ff2ec1dd00121e599cbccd98dce463d1cdbf22c113d0a924969ffe78eac559	0
bccsp/factory/swfactory.go	2196	d08d9a5961a76c2ded743dde029d20da7644a6fb329e3f27e95db53f2652360c	0
bccsp/hashopts.go	1861	29171b614a86730605c063fe1b05ca4abae7dbdb55765d15eb528371fc97f180	0
bccsp/keystore.go	1197	07eea30ab3a8cc6f2b94fb686cb158b64c27d95339c1603fe4153531a1e5b39c	0
bccsp/opts.go	10150	ed73949f83ce35e62e2d29bc2263b6b4e730e24f7e7e81582a32fd6b14c0c606	0
bccsp/pkcs11/conf.go	1330	1cc57eb00ecafb06d5c6d4ff8dea228eae6ed6b50f935c71451d5fa62d2bec06	0
bccsp/pkcs11/ecdsakey.go	2564	41b35f8b4e01edc52825fc73e1041ce296e6fc1d979622d7b437424f72160bd9	0
bccsp/pkcs11/pkcs11.go	27875	98a15788cd8cdca171d2c662f6e2eece206a5d8adc91994f75ddd67ffc6bbd4b	0
bccsp/rsaopts.go	1866	2a90bd507977049cd6a43eca3597569935c3f114c227ba1000dcf1113f3c4543	0
bccsp/signer/signer.go	2574	1806b0e0b2a4dc08d34a1285eeef7349b6a20bcdfa555363cdee4bf281bea41d	0
bccsp/sw/aes.go	5947	25e0c5cb3d867fa8c92f38a5e42c346538da18798f229934c0c00755fc4c057d	0
bccsp/sw/aeskey.go	1741	08b6095ec603d320e7839a36859404474eb443e640dc12c06718bb7cc5931dea	0
bccsp/sw/conf.go	1593	abcd19356cf759f0688ee38889e90bfd8ba8fc6a3b4608eb50653dde9b13e177	0
bccsp/sw/dummyks.go	1532	f6d1bc4e721f58f5471f5aef3659e3ba46fc822306d0093d556ede2df66dcb10	0
bccsp/sw/ecdsa.go	2224	f0d2594a515e856b310d5ff9ab4169d5c0a0f61fc24942e3d46e2500f928e02b	0
bccsp/sw/ecdsakey.go	3024	194f71f7e0d952ec3b87a50b5287d7bd139cb41941d75cd260e2e54ab8b01ff3	0
bccsp/sw/ed25519.go	1208	071a175c9ce2398e741a262d24325b73260f7d87ea7be622eebb9f505880d580	0
bccsp/sw/ed25519key.go	2555	5fd0b8364abab17d1378f6aba697ce2ffc0f47138f2d02056b75dd9178e9b158	0
bccsp/sw/fileks.go	11165	a0c92a6cfdcba39bd63cd9ce550b20f3a59567e53507468644c3f82330caa6db	0

bccsp/sw/hash.go	923	b5b4f1a45b5e2f3fd10d4e505ffa08ed1a087d6e9336af29357e11ceb10b4e0c	0
bccsp/sw/impl.go	9978	e86e5b101dcb085adbd489099d8566e0207e30265537f77ece063d71d0201eb0	0
bccsp/sw/internals.go	3357	855ba254295b233c474266690c944f8b5f0c425070fdcee8e5b72652d8d211d5	0
bccsp/sw/keyderiv.go	4131	fb6673a63d52fa58cdb93b3e75aeb88faf2c2c91ac3e29de9acd7f519f8ac418	0
bccsp/sw/keygen.go	1545	51177e8239b008e100259033f1075396fac4711b13d71a27d7fe1fd1697db1b4	0
bccsp/sw/keyimport.go	6380	573e238eea23ae85576a43554db460121e1ac8666fafebb84138326deedd0f25	0
bccsp/sw/keys.go	11408	cd0a974ae8900553a3f4828afcbbdd30cd479b75a51bc1723d369220e739fe3c	0
bccsp/sw/new.go	5934	fb412076a0df7836edb4a8823024134f0151607cbc8d6ac6ce27597d1cdb1c2e	0
bccsp/sw/rsa.go	1552	e3112f8c38642c19ebbb5a18c26ce471ec9fd56fae78909c4ff216d5216d2c42	0
bccsp/sw/rsakey.go	2754	93838157dd6a4eca5ecf566fa365064df1afa290b08829ae97074ccfcd473de1	0
bccsp/utils/ecdsa.go	2568	4d39accb8acba4c0041942b921c3a501e15926b19900e6ca8e5980bd5c863660	0
common/flogging/core.go	3184	ae5cbaeec703dceb5601ac4e5cf77c6c725ce02c6b1edc537d7b3bb9fc843196	0
common/flogging/fabenc/color.go	530	d32969c9dc25fa3a22acc28f2c289657433b5c06f154be3348cc8766d5ffa933	0
common/flogging/fabenc/encoder.go	3957	93ee108897c7c6dba06842927ae1701dc4dffaaea6983e0dd6f845702001cf36	0
common/flogging/fabenc/formatter.go	8861	bf6d5b76ee67a4d227fca209207f85ad41fc627de4450f8bd0f601f961d63c32	0
common/flogging/floggingtest/logger.go	5246	62d66823c89e6e9d93e5fe2712517ae65fa801885d0c4a4393931aae2f89568e	0
common/flogging/global.go	1876	d7b8402851c29934800911208966e44d8ff8c64056c7b6bdbaf91f9460931128	0

common/flogging/httpadmin/fakes/logging.go	4440	01ef7790f685bbd3884afc1e80e8e562ed54aa773b4f902bac5245e56c459e8f	0
common/flogging/httpadmin/spec.go	1860	73f3e5e5344a6f6efeb3e669f496e9010906eb2d0a36f663b266a695250a4483	0
common/flogging/levels.go	1565	aff982968eb7cf75d91c3320aba1889201187cfe4da94f10b022ebcd7eaba20a	0
common/flogging/loggerlevels.go	4891	d9ae711ddec04454937fb5c0197e90a6602af38f3dfef6d10fc27a2fcdd0a9fe	0
common/flogging/logging.go	5924	c24bcf1be2c7c246a8463974d5abff4b2768bf2bbdfbe4dd86ae706b832f044a	0
common/flogging/metrics/observer.go	1249	a9a75918b8b3ae94742ef39785c1ca947153374ca58abdd83368fb50623aa79d	0
common/flogging/zap.go	5211	0e1f06d541eed96918d47136c5d09d7a90ff64929ec223017446de85ccb929c3	0
common/metrics/cmd/gendoc/main.go	1641	0c47cc05ac4e472b424c81945358ac1cc07d21f2304610e7825ccd5ae96b7305	0
common/metrics/disabled/provider.go	947	cf31485e6ce257ecc591b9a79336a2e0fac078d4aa54565b9a3ae5c6670c4f7c	0
common/metrics/gendoc/options.go	4968	5d004d5bdc385491ccf26e12fa1e5fdd7b43348acc7b9ab16a5b53e7e3ca184b	0
common/metrics/gendoc/table.go	8899	893d08a5996bbc4a7d21c649ab44259d9d402373e2c56654ef9395cab9160b54	0
common/metrics/metricsfakes/counter.go	3624	4768783f78325d4c6590f4cd719a53ab3c61dd906a0dfc978f75622d7b1b9969	0
common/metrics/metricsfakes/gauge.go	4434	4a82092ced01547f3ad4d6abf1028a7afd1542ae133909d3ba3cf726e59d4b5c	0

common/metrics/metricsfakes/histogram.go	3774	ba7ca9aa1b4df78de4f5528e97e12708b9a42332dabd1ea9cd385abd5971697f	0
common/metrics/metricsfakes/provider.go	7331	482bdc8f1de53bdcb3553c204a0cc0eae65d0478aa209c563d1e26c9de637ed1	0
common/metrics/namer/namer.go	3213	269783a19cec93d1023653757d9880d2b688d2e8736663a002c1af0ac10e7899	0
common/metrics/prometheus/provider.go	1720	af19837203d17535cdb8f0f73e3b31ee64b59f74c1aff603f2c787f1853638ef	0
common/metrics/provider.go	6241	2e302c1f8f336fdeaa892bbb046dbfb96d9e535004d0fe964785946e495cda8c	0
common/metrics/statsd/goruntime/collector.go	4381	7a72e9edfd4a4c7073d7890e38ec4a4994b187722e5429742b6893825db17e06	0
common/metrics/statsd/goruntime/metrics.go	3003	c361fb329763628f27925de1d41e8ae1084293a6c30ddf57109f9fd690ee45e1	0
common/metrics/statsd/provider.go	2835	cc796d42a6f65cf46919a3ba217bcca739d42e0aaae470c258d4cfadae0bb219	0
go.mod	1877	f0b24cd1313a8979feb1f5e48251b2ee4ad8ad0a6c993c8f52ea581c39e41e5d	0
healthz/checker.go	4918	bec07fbe52e60910b0fc85149ea4f9b5e502f7e29ea3ad37a61f2dbf795a8d28	0
tools/go.mod	348	7b34cca11276a6eb65b446751aa74d984045b3e2fee742b56c25cfae62b3d5b5	0